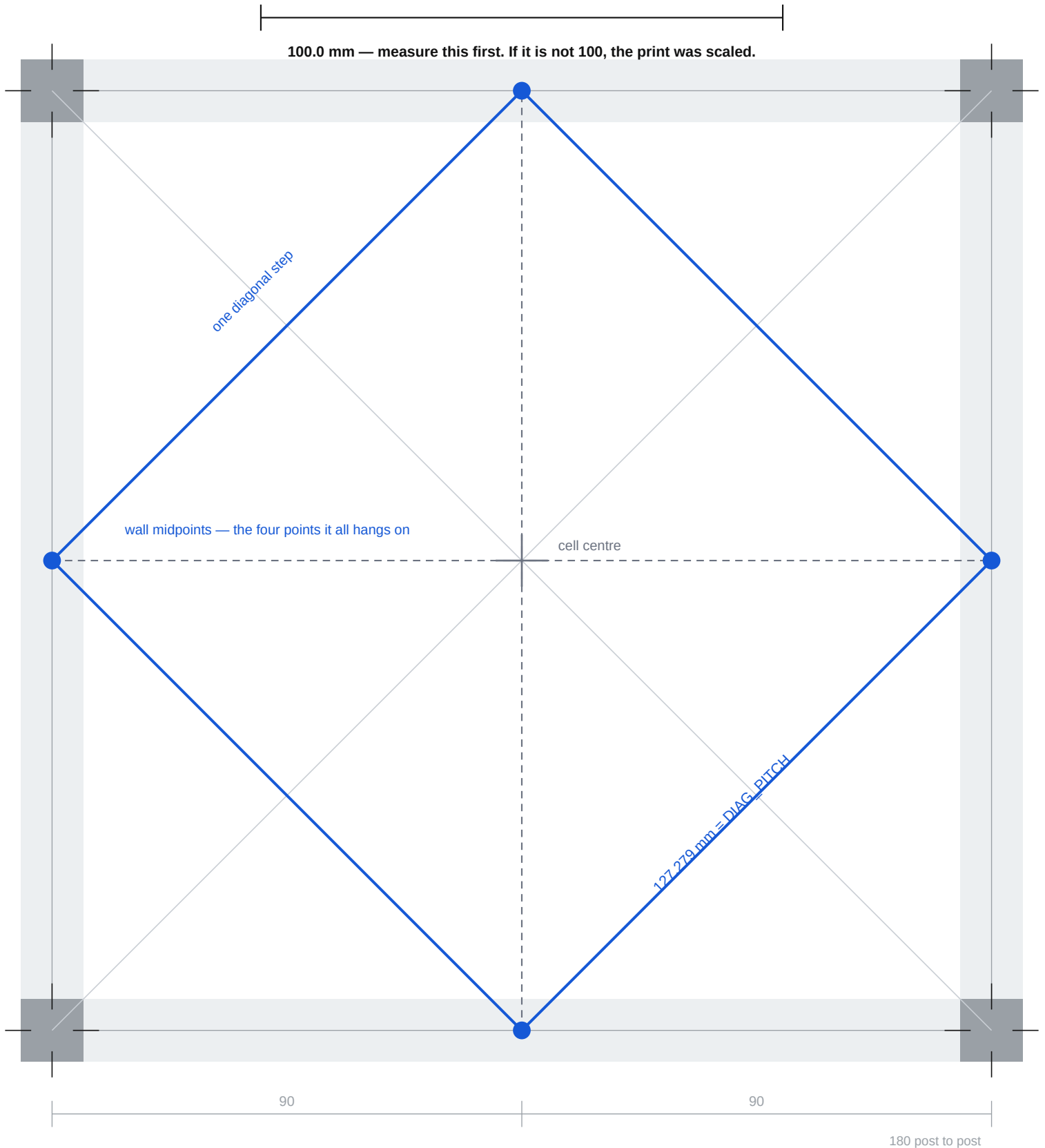


E4 — cell template, 1:1

lay it in a cell, square it to the posts, mark through the four midpoints

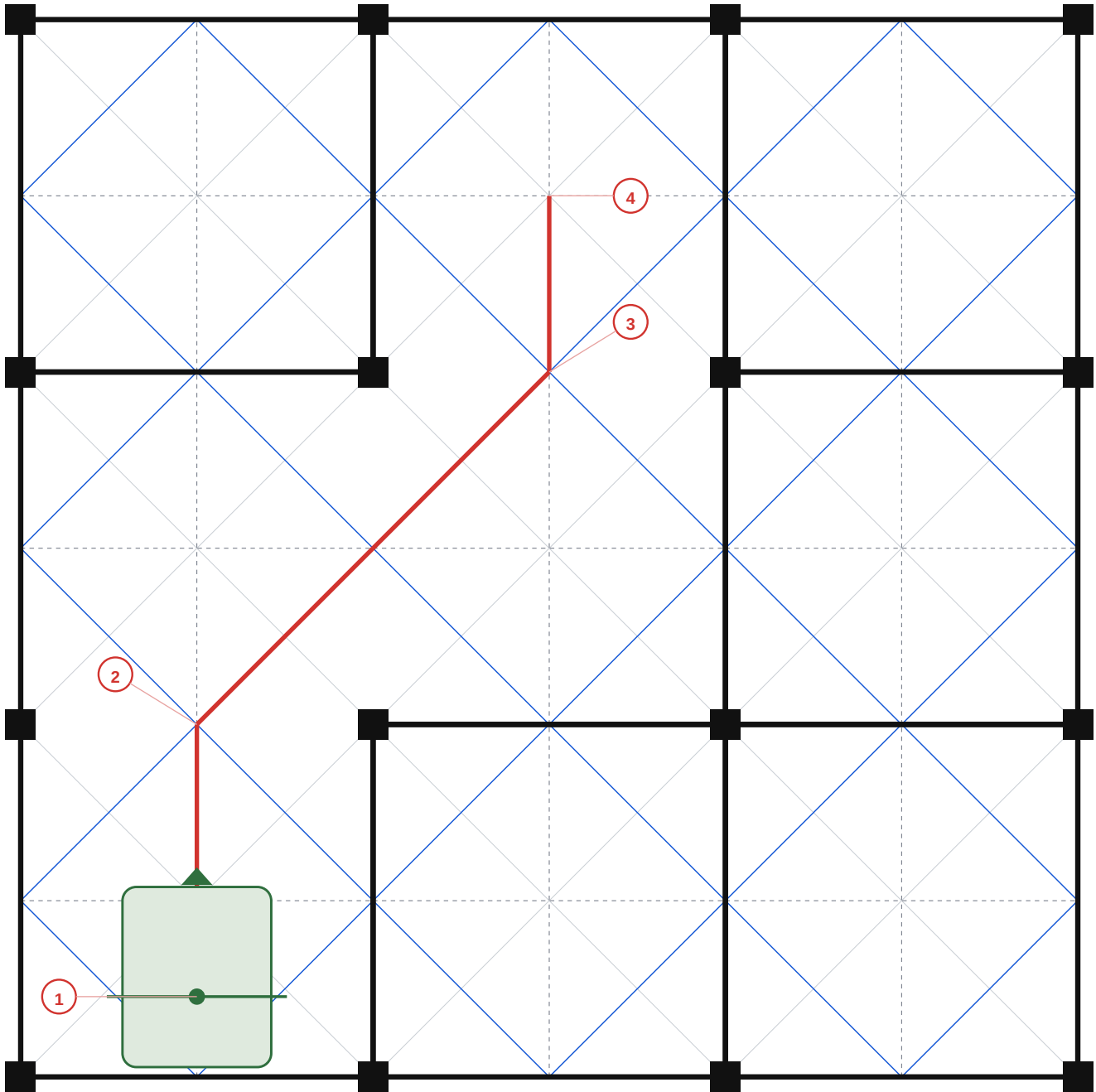


- diagonal running line**
joins the four wall midpoints. The lane she is on between an SD45 and a DS45.
- orthogonal centrelines**
the straight lanes — 90 mm from the boundary, 84 mm clear of a wall face.
- corner to corner**
post to post. For checking squareness and 45 by eye — she never drives on it.

To mark it out: find each wall midpoint (90 mm from either post) and join the four. That diamond passes a post on every side without touching one, which is the whole point of it. The cell centre is NOT on it — that is what the 45s have to bridge.
E4 · bench diagnostics

Where she goes — right-hand 45

one 3×3 section — a picture of where she goes. Not to scale: page 1 is the one you measure from.



bench-3x3-45minR — right hand onto the diagonal, left hand off it

- ① Put her down here, backed square to the wall, facing north. Row 7 (SD45R), approach 1 cell.
- ② SD45_R joins the diagonal HERE — the wall midpoint, 90 mm before the cell centre above it.
- ③ One diagonal step later, 127.279 mm, DS45_L puts her back on a straight lane.
- ④ She stops on this cell centre. Watch: did she land ON the blue line, and leave it square?

Read this off the drawing before you trust the table

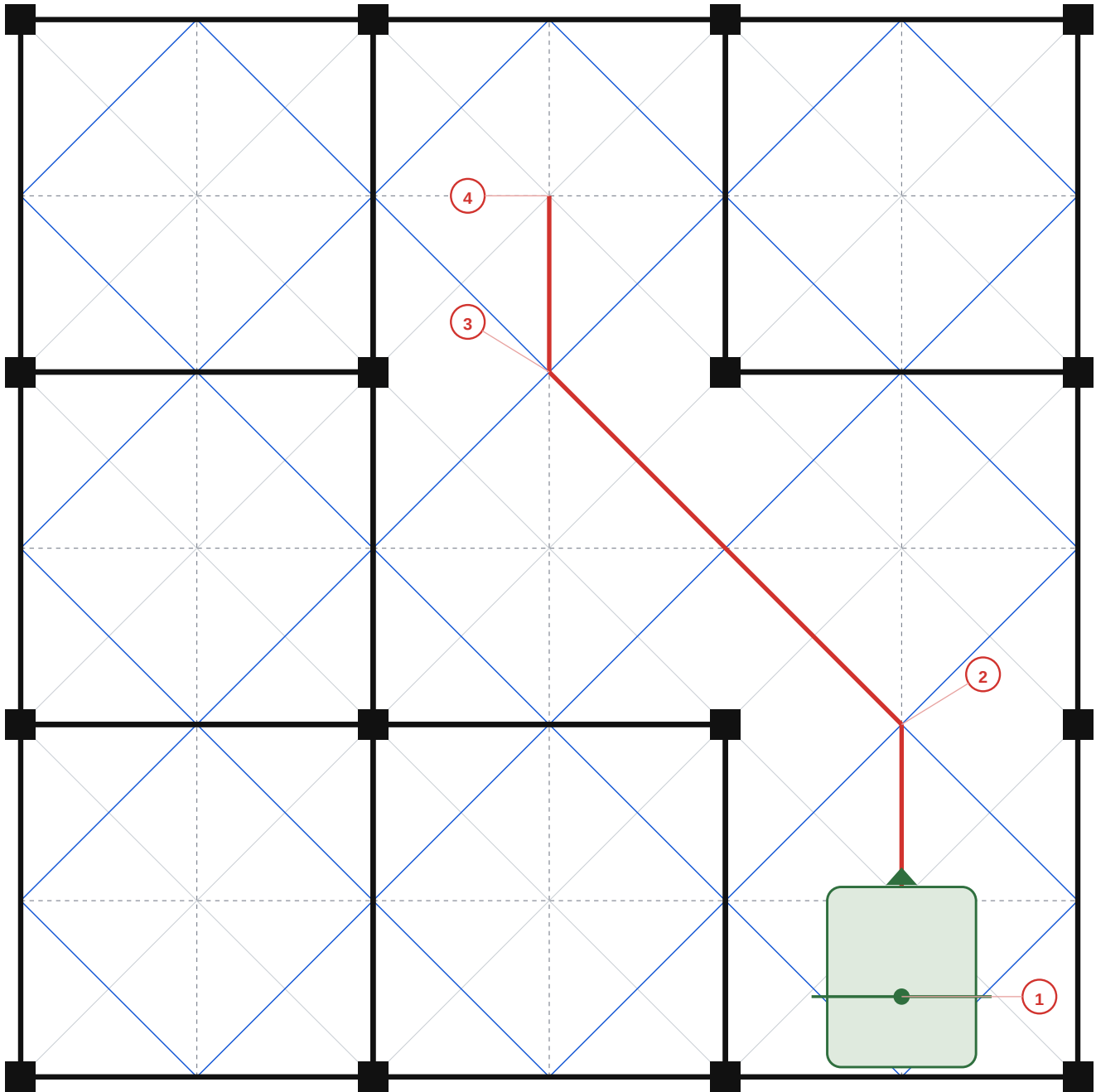
The straight lane and the diagonal cross at the WALL MIDPOINT — 90 mm before the centre of the cell the SD45 is booked in. entry_offset is measured back from that CELL CENTRE, so an SD45 needs $90 + R \cdot \tan(22.5^\circ)$ to be tangent to both lanes.

At $\omega 95$ that is $90 + 75 = 165$ mm. The table holds 120, which starts the arc 30 mm before the crossing instead of 75.

DS45 and DD90 turn AT their lattice point, so those need $R \cdot \tan(\theta/2)$ with no 90 added — 75 and 63. Only SD45 carries the half cell.

Where she goes — left-hand 45

one 3×3 section — a picture of where she goes. Not to scale: page 1 is the one you measure from.



bench-3x3-45minL — the mirror. Left hand onto the diagonal, right hand off it

- ① Put her down here — the bottom-RIGHT cell, backed to the wall, facing north. Row 6 (SD45L).
- ② SD45_L joins the diagonal at the wall midpoint, exactly as its mirror does.
- ③ DS45_R leaves it one step later. Same distances, other hand.
- ④ Same finishing cell as the right-hand rig. Any difference between the two is the mouse.

Read this off the drawing before you trust the table

The straight lane and the diagonal cross at the WALL MIDPOINT — 90 mm before the centre of the cell the SD45 is booked in. entry_offset is measured back from that CELL CENTRE, so an SD45 needs $90 + R \cdot \tan(22.5^\circ)$ to be tangent to both lanes. At ω 95 that is $90 + 75 = 165$ mm. The table holds 120, which starts the arc 30 mm before the crossing instead of 75. DS45 and DD90 turn AT their lattice point, so those need $R \cdot \tan(\theta/2)$ with no 90 added — 75 and 63. Only SD45 carries the half cell.